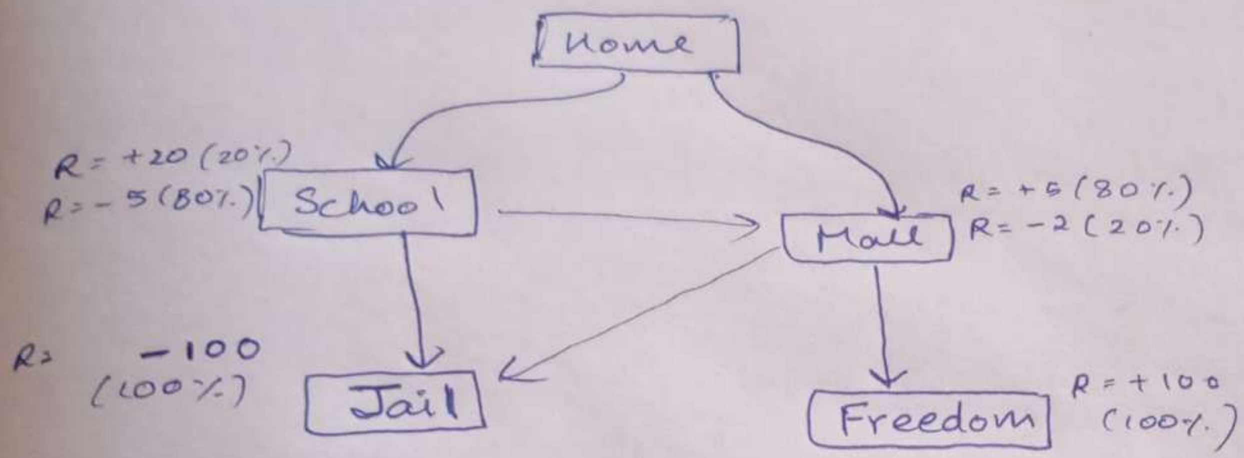

At Home
$\boxed{A_1}$ ⇒ 70% Home → Sch 30% Home ⇒ Mall
$\boxed{A_2}$ ⇒ 70% Home → Mall 30% Home ⇒ Sch

At Mall
$\boxed{A_3}$ → 70% Mall → Freedom
30% Mall → Jail

At school
$\boxed{A_4}$ → 70% School → Jail
30% School → Mall

Q - Table k=0

	A_1	A_2	A_3	A_4
Home	0	0	—	—
School	—	—	—	0
Mall	—	—	0	—
Jail	—	—	—	—
Freedom	—	—	—	—

$\gamma = 0.9$

k=1

$$Q_1(\text{Home}, A_1) = E\left(R(s' \mid \text{Home } A_1) + \gamma \max_a Q(s', a)\right]$$

$$= 0.7\left[20\left(\tfrac{1}{5}\right) - 5\left(\tfrac{4}{5}\right) + \gamma Q(\text{school}, A_4)\right] \Rightarrow \text{School}$$

$$+ 0.3\left[5\left(\tfrac{4}{5}\right) - 2\left(\tfrac{1}{5}\right) + \gamma Q(\text{Mall}, A_3)\right] \Rightarrow \text{Mall}$$

$$= 1.08$$

$$Q_1(\text{Home}, A_2) = \underbrace{0.7\left[3.6 + \gamma(0)\right]}_{\text{Mall}} + \underbrace{0.3\left[0 + \gamma(0)\right]}_{\text{School}}$$

$$= 2.52$$

$$Q_1(\text{Mall}, A_3) = \underbrace{0.7\left[100 + \gamma(0)\right]}_{\text{Freedom}} + \underbrace{0.3\left[-100 + \gamma(0)\right]}_{\text{Jail}}$$

$$= 40$$

$$Q_1(\text{School}, A_4) = \underbrace{0.3\left[3.6 + \gamma(0)\right]}_{\text{Mall}} + \underbrace{0.7\left[-100 + \gamma(0)\right]}_{\text{Jail}}$$

$$= -68.92$$

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Q Table $k=1$

	A_1	A_2	A_3	A_4
Home	1.08	2.52	–	–
School	–	–	–	-68.92
Mall	–	–	40	–
Jail	–	–	–	–
Freedom	–	–	–	–

$k=2$

$$Q_2(\text{Home}, A_1) = 0.7\underbrace{[0 + 0.9(-68.92)]}_{\text{School}} + 0.3\underbrace{[3.6 + 0.9(40)]}_{\text{Mall}} = -31.54$$

$$Q_2(\text{Home}, A_2) = 0.7\underbrace{[3.6 + \gamma 0.9(40)]}_{\text{Mall}} + 0.3\underbrace{[0 - 0.9(68.92)]}_{\text{School}} = 9.11$$

$$Q_2(\text{Mall}, A_3) = 0.7\underbrace{[100 + \gamma(0)]}_{\text{Freedom}} + 0.3\underbrace{[-100 + \gamma(0)]}_{\text{Jail}}$$

$$= 40 \text{ [convergent]}$$

$$Q_2(\text{School}, A_4) = 0.3\underbrace{[3.6 + \gamma 0.9 \times 40]}_{\text{Mall}} + 0.7\underbrace{[-100 + 0.9(0)]}_{\text{Jail}}$$

$$= -58.12$$

Q-Table
$k=2$

	A_1	A_2	A_3	A_4
Home	-31.54	9.11	–	–
School	–	–	–	-58.12
Mall	–	–	40	–
Jail	–	–	–	–
Freedom	–	–	–	–

[For now A_2 is the best action to take at Home]